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Cancer incidence rates among South Asians in four geographic regions: India, Singapore, UK and US

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Background Data are limited regarding cancer incidence among Indians residing in different geographic regions around the world. Examining such rates may provide us with insights into future aetiological research possibilities as well as screening and prevention.

Methods Incidence rates for all cancers combined and 19 specific cancers were obtained for India from Globocan 2002, for Indians in Singapore from Cancer Incidence in Five Continents (VIII), and from national data sources for South Asians (SA) in the United Kingdom (UK) and for Asian Indians/Pakistanis (AIP) and whites in the United States (US).

Results We observed the lowest total cancer incidence rates in India (111 and 116 per 100 000 among males and females, respectively, age-standardized to the 1960 world population) and the highest among US whites (362 and 296). Cancer incidence rates among Indians residing outside of India were: intermediate Singapore (102 and 132), UK (173 and 179) and US ranges 152–176 and 142–164. A similar pattern was observed for cancers of the colorectum, prostate, thyroid, pancreas, lung, breast and non-Hodgkin lymphoma. In contrast, rates for cancers of the oral cavity, oesophagus, larynx and cervix uteri were highest in India. Although little geographic variability was apparent for stomach cancer incidence, Indians in Singapore had the highest rates compared with any other region. The UK SA and the US AIP appear with adopt the cancer patterns of their host country.

Conclusion Variations in environmental exposures such as tobacco use, diet and infection, as well as better health care access and knowledge may explain some of the observed incidence differences.

Keywords Cancer incidence, India, Asian Indians, South Asians, Singapore Indians

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Introduction

Dramatic international variations exist in cancer incidence rates.[1] While recent efforts have estimated cancer incidence rates for India as a whole,[2] cancer incidence among Asian Indians in different areas of the world have not been compared to any great extent.[3] Only in Singapore have Asian Indians been specifically identified in the cancer registry. Seminal observations on Japanese and other Asian Americans led to epidemiologic research in the West examining dietary and other factors associated with cancer risk.[4–6] Our understanding of cancer epidemiology, therefore, could be significantly broadened, including revealing factors requiring investigation, by studying other diverse populations.[7]

Asian Indians represent one-sixth of the world population, with a global total of over one billion and approximately 20 million living outside India. The population of Asian Indians/Pakistanis (AIP) in the United States (US) was 1.84 million in 2000 and rising.[8,9] In the United Kingdom (UK), the South Asian (SA) population was 2.27 million in 2001.[10]

Data are emerging regarding cancer rates among Asian Indians in the US and the UK.[11–17] We present estimated national incidence rates in India, rates among Indians and other SA residing in Singapore, the UK and the US and rates among US whites.

Methods

Data sources

India

National estimates of cases and rates for India as a whole were obtained from the IARC Globocan 2002 data set.[2] These estimates were derived using data for 1993–97 from the nine regional population-based registries (Ahmedabad, Bangalore, Chennai, Delhi, Karunagappally, Mumbai, Nagpur, Poona and Trivandrum) included in Volume VIII of Cancer in Five Continents[3] and took into account regional and urban/rural variations in rates.

Singapore

Data for 1993–97 incident cases of cancer and rates for the Singapore Cancer Registry were abstracted from Cancer Incidence in Five Continents, Vol. VIII.[3] The Singapore registry included an Indian subcontinent population of about 220 000 (includes Indians, Pakistanis and Sri Lankans).

United Kingdom

Rates for UK SA (includes Indians, Pakistanis and other SA) were derived using cancer registry and population census data. The National Cancer Intelligence Centre (NCIC) of the Office for National Statistics receives notification of cancer registrations for England and Wales from 10 regional cancer registries. As ethnicity information on cancer cases submitted to the NCIC, including persons of SA origin, is incomplete or unavailable, a validated computerized name algorithm, South Asian Names and Group Recognition Algorithm (SANGRA)[18] was used to identify SA cancer cases diagnosed in the period 1999–2001. SANGRA identifies persons of SA origin by matching the names in the NCIC data to SA names contained in its dictionaries. The algorithm has been shown to have over 89–96% sensitivity and 94–98% specificity when validated against self-described ethnicity from a number of localities in England.[18] To improve specificity further, all names identified as being SA were visually inspected and excluded if they were common to other ethnic groups. As a check on sensitivity, 1000 randomly selected names were visually inspected to assess the percentage not detected by the algorithm. Three names, out of a random sample of 1032 names, that had not been picked by SANGRA were found to be of SA origin, suggesting that 0.3% of cases may have been missed.

Population estimates of UK SA were available from the 2001 Census for England and Wales. They included all persons self-described as Asian or Asian British, and of specifically East African Asian or SA origin ($n = 2.27$ million).

United States

Rates for AIP were derived using cancer registry and population census data. We obtained data for cancer cases diagnosed among individuals of self-reported AIP race/ethnicity during the time interval 1999–2001 from the Surveillance, Epidemiology, and End Results Program (SEER)[19] of the National Cancer Institute (using *SEER*Stat* software program). The SEER 11 database combines data from cancer registries in Atlanta, Connecticut, Detroit, Hawaii, Iowa, Los Angeles County, New Mexico, San Francisco-Oakland, San Jose-Monterey, Seattle-Puget Sound and Utah. As SEER 11 covers a small proportion of the total US AIP population [19% (352 573) of the US AIP population (1.84 million)], we supplemented SEER 11 data with corresponding self-reported ethnicity data provided by state registries in California,[20] New York,[21] New Jersey,[22] Texas,[23] Florida,[24] Pennsylvania[25] and Michigan.[26] State registry data were used instead of SEER data when there was an overlap. State registry data in conjunction with SEER enabled us to cover 1.24 million of the US AIP population.

US AIP population data were obtained from the 2000 US Census.[8] The US 2000 census forms allowed persons to choose more than one ethnicity, resulting in two population values (lower and upper estimates): the lower bound represents persons identifying themselves as Indian or Pakistani only, and the upper bound, these same persons plus those who selected another ethnicity in addition to Indian or Pakistani. The number of persons indicating Indian or Pakistani alone or in any combination was 15.3% higher than the number indicating Indian or Pakistani alone. We therefore derived lower and upper estimates for cancer incidence rates for US AIP, using cases as numerators and the two population estimates as denominators. Cancer incidence rates for US whites were obtained from SEER 11.[19]

Data analysis

Using data from each population, we derived cancer incidence rates for all cancers, excluding non-melanoma skin cancers, and for 19 specific primary cancers: oral cavity and pharynx; oesophagus; stomach; colorectum; liver; pancreas; larynx; lung and bronchus; breast; cervix uteri; corpus and uterus, not otherwise specified (NOS); ovary; prostate; urinary bladder; kidney and renal pelvis; brain and nervous system; thyroid cancers; non-Hodgkin lymphoma (NHL) and leukaemia. Incidence rates were age-standardized to the 1960 Segi world population (IARC, WHO).[2] We deemed rates unstable if the number of cancer cases was less than 12 and the rate was not shown. Standard errors based on the numbers of cases and rates were estimated[27] for all the rates except those for India, for which the numbers are estimated for the entire country, and actual counts are not available.[2]

Rate ratios (RR) were calculated, comparing incidence rates of each population (Singapore Indians, UK SA, US AIP upper and lower bounds and US whites) with rates in India (referent group) for the total and for 19 specific primary cancers. Age-incidence curves were plotted on semi-logarithmic graphs using 10-year age-specific rates for each region except India, which used the 20-year or larger age-specific rates available from Globocan 2002.[2]

Results

Overall rates

Overall age-adjusted cancer incidence rates among both males and females were lowest in India and Singapore Indians, intermediate in UK SA and US AIP and highest in US whites (Table 1). Notably, the rates for all cancers combined were higher among females than males in India and Singapore Indians. Only in the US were rates distinctly higher among males than females among both AIP and whites.

Leading cancers in each population

The five most common cancers among Indian males were cancers of the oral cavity and pharynx, lung, oesophagus, larynx and stomach. Similarly, Singapore Indian males had high rates of oral cavity and pharynx, lung and stomach cancers, but prostate and colorectal cancers replaced oesophagus and larynx cancers in the top-five malignancies. The patterns among UK SA and US AIP males were more similar to US white males than their counterparts in India or Singapore, as the four most common malignancies were prostate, lung, and colorectal cancers and NHL. The fifth most common cancer among UK SA and US API was leukaemia, in contrast to urinary bladder cancer among US white men.

Among females, cervix uteri cancer was among the top-five cancers only in India and Singapore. Breast cancer was among the five most frequent cancers diagnosed in all the populations studied. Interestingly, ovarian cancer was one of the top-five cancers among all four Indian populations investigated but not among US white females. Oral cavity and pharynx cancer was among the top five for Indian females and Singapore Indians, while oesophagus cancer was only in the top five among Indian females. In contrast, colorectal cancer was among the five leading cancers in all groups except Indian females. Corpus uteri cancer was among the top-five cancers in UK SA, US AIP and US whites.

Comparison of rates across populations

Among males, the rate for all cancers combined was similar in India and among Indians in Singapore (Figure 1, panel A). The overall rates among SA in the UK and the US were 50–75% higher, and among US whites more than three times those in India. Relative to rates in India, prostate cancer rates were more than 20 times higher in US whites, more than 10 times higher in US AIP, seven times higher among UK SA and twice as high among Singapore Indians. These were the largest relative differences apparent. Colorectal cancer rates were almost eight times higher in US whites, about three times higher among US AIP and UK SA than in India, and almost twice as high among Singapore Indians. Similar, though somewhat less dramatic, gradients were apparent for urinary bladder, kidney and renal pelvis, lung and thyroid cancers and leukaemia. In contrast, rates for oral cavity and pharynx, oesophagus and larynx cancers all were highest in India, with the rates among SA living in the UK or the US lower than even US white rates. Stomach cancer rates were highest among Singapore Indians and varied little among the other population groups.

Among females, the total cancer rate was lowest in India, about 26% higher in Singapore, 36–57% higher among US AIP, 72% higher among UK SA and >180% higher among US whites (Figure 1, panel B). The largest relative differences were for lung cancer, with the RR relative to India being about three for Singapore Indians, four among UK SA and US AIP and 16 for US whites. Colorectal cancer RRs were around four among Indians in Singapore, the UK and the US, and more

Table 1 Cancer incidence in India, Singapore Indians, UK SA, US Asian Indians and Pakistanis and US whites

Cancer	India[a] 1993–1997		Singapore Indians 1993–1997			UK SA 1999–2001			US A/P 1999–2001					US whites 1999–2001		
										Rates[b] and Standard errors						
	Counts	Rates[b]	Counts	Rates[b]	Standard error[c]	Counts	Rates[b]	Standard error[c]	Counts	Lower bound[d]	Standard error[c]	Upper bound[d]	Standard error[c]	Counts	Rates[b]	Standard error[c]
A. Males																
All cancers	404 309	99.0	649	101.5	4.0	4919	172.9	2.5	2508	151.6	3.0	175.8	3.5	205 440	362.4	0.8
Oral cavity and pharynx	92 808	22.9	67	10.6	1.3	176	6.1	0.5	105	5.7	0.6	6.6	0.6	5861	10.7	0.1
Oesophagus	29 652	7.6	20	3.2	0.7	93	3.4	0.4	30	2.0	0.4	2.4	0.4	2925	5.1	0.1
Stomach	22 650	5.7	57	9.0	1.2	180	6.4	0.5	74	4.6	0.5	5.4	0.6	3842	6.4	0.1
Colon and rectum	19 508	4.7	56	8.3	1.1	484	17.3	0.8	230	13.0	0.9	15.0	1.0	22 211	37.1	0.2
Liver	9 153	2.3	50	7.9	1.1	186	6.6	0.5	68	4.2	0.5	4.9	0.6	2764	4.3	0.1
Pancreas	5 711	1.4	16	2.3	0.6	162	6.0	0.5	57	3.5	0.5	4.1	0.5	4474	7.6	0.1
Larynx	24 216	6.2	33	5.0	0.9	73	2.7	0.3	29	1.9	0.4	2.3	0.4	2441	4.5	0.1
Lung and bronchus	35 495	9.0	68	10.0	1.2	617	22.4	0.9	252	16.8	1.1	19.5	1.2	27 425	47.1	0.3
Prostate	16 789	4.6	65	9.9	1.2	891	33.7	1.1	696	47.4	1.8	54.9	2.1	62 398	113.1	0.5
Urinary bladder	12 444	3.2	33	5.4	0.9	186	7.0	0.5	105	6.8	0.7	7.9	0.8	14 170	23.1	0.2
Kidney and renal pelvis	4 738	1.2	17	2.8	0.7	128	4.4	0.4	82	4.7	0.5	5.4	0.6	6268	11.4	0.1
Brain and nervous system	12 150	2.6	14	2.7	0.7	169	5.4	0.4	85	4.2	0.5	4.9	0.5	3278	6.5	0.1
Thyroid	4 361	1.0	–[f]	–	–	38	1.2	0.2	52	2.0	0.3	2.3	0.3	1779	3.3	0.1
Non-Hodgkin lymphoma	13 900	3.2	19	3.1	0.7	347	11.3	0.6	156	8.3	0.7	9.6	0.8	9272	16.3	0.2
Leukaemia	15 062	3.1	28	4.9	0.9	282	9.3	0.6	141	7.4	0.6	8.6	0.7	6153	11.4	0.1

B. Females

All cancers	447 592	104.4	554	131.7	5.6	5119	179.2	2.5	2464	142.2	2.9	164.4	3.3	198 585	295.6	0.7
Oral cavity and pharynx	39 849	9.6	30	7.7	1.4	174	6.1	0.5	62	3.4	0.4	3.9	0.5	2914	4.2	0.1
Oesophagus	20 805	5.1	–	–	–	86	3.3	0.4	28	1.9	0.4	2.2	0.4	3842	1.2	0.0
Stomach	11 743	2.8	23	6.0	1.3	71	2.6	0.3	44	2.8	0.4	3.2	0.5	2423	3.0	0.1
Colon and rectum	13 555	3.2	53	14.9	2.0	333	12.1	0.7	168	10.2	0.8	11.8	0.9	21 842	26.6	0.2
Liver	4 477	1.1	–	–	–	72	2.7	0.3	25	1.7	0.3	2.0	0.4	1371	1.5	0.0
Pancreas	3 506	0.8	–	–	–	102	3.9	0.4	35	2.3	0.4	2.7	0.5	4485	5.5	0.1
Larynx	3 157	0.8	–	–	–	11	0.4	0.1	–	–	–	–	–	660	1.1	0.0
Lung and bronchus	8 046	2.0	17	5.4	1.3	189	7.3	0.5	113	7.3	0.7	8.5	0.8	23 495	33.0	0.2
Breast	82 951	19.1	167	36.7	2.8	1894	64.6	1.5	918	50.6	1.7	58.2	1.9	64 645	101.7	0.4
Cervix uteri	132 082	30.7	35	8.2	1.4	145	4.6	0.4	81	4.7	0.5	5.4	0.6	3926	7.0	0.1
Corpus and uterus, NOS[e]	6 937	1.7	31	6.9	1.2	265	9.8	0.6	139	8.3	0.7	9.6	0.8	11 786	18.7	0.2
Ovary	21 146	4.9	45	9.2	1.4	278	9.5	0.6	135	7.3	0.6	8.4	0.7	6820	10.6	0.1
Urinary bladder	3 031	0.7	–	–	–	49	1.9	0.3	35	2.4	0.4	2.8	0.5	4732	6.0	0.1
Kidney and renal Pelvis	2 129	0.5	–	–	–	65	2.3	0.3	30	1.8	0.3	2.0	0.4	3828	5.8	0.1
Brain and nervous system	7 530	1.6	–	–	–	111	3.9	0.4	46	2.5	0.4	2.9	0.4	2468	4.4	0.1
Thyroid	8 686	1.9	22	3.5	0.7	120	3.6	0.3	130	6.2	0.5	7.1	0.6	5174	9.7	0.1
Non-Hodgkin lymphoma	7 389	1.7	12	2.5	0.7	247	8.7	0.6	89	5.5	0.6	6.4	0.7	8	11.0	3.9
Leukaemia	9 778	2.1	–	–	–	203	7.2	0.5	85	4.9	0.5	5.8	0.6	4610	7.2	0.1

[a]As estimated by Globocan 2002, IARC; standard errors not calculated because actual national India counts are not available.
[b]Rates per 100 000, age-adjusted to the 1960 Segi world population.
[c]Standard error estimated as the rate/square root(count) (3).
[d]Lower and upper bound estimates for US AIP cancer incidence rates are presented since the US 2000 Census provides two values for the US AIP population (Asian Indian or Pakistani alone, or in combination with another ethnicity).
[e]Not otherwise specified.
[f]Dash indicates statistic not shown, rate based on less than 12 cases.

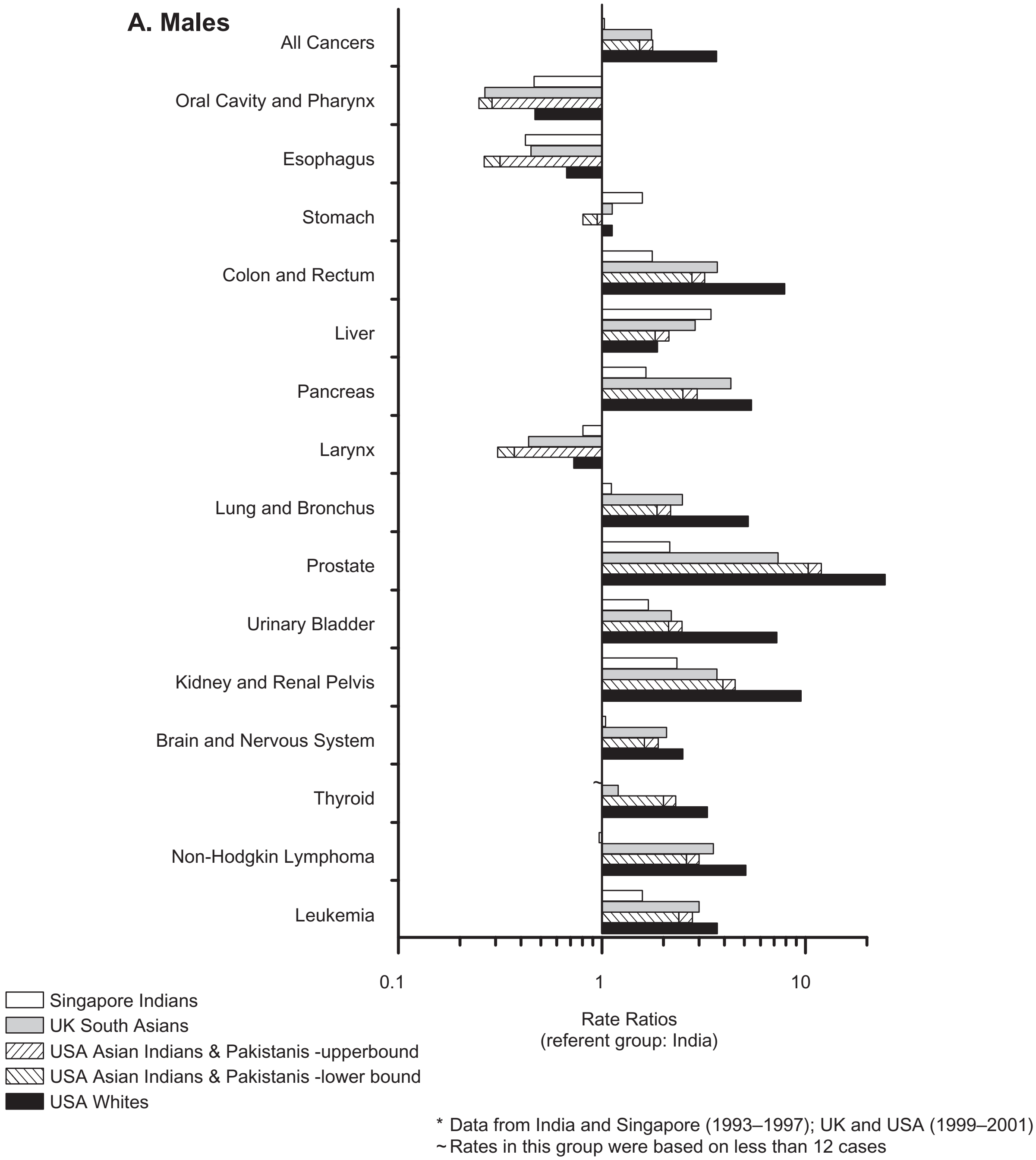

Figure 1 Rate ratios of directly age-adjusted (world standard) cancer incidence rates among three Indian groups (residing in Singapore, UK or US) and US whites, relative to rates in India: **A**. Males; **B**. Females

than eight for US whites. Rates for cancers of the breast, corpus uteri, urinary bladder, kidney and renal pelvis and thyroid showed similar, although less dramatic, gradients, with rates lowest in India, intermediate among Indians elsewhere, and highest among US whites. As with the rates among males, Indian females had the highest rates of oral cavity/pharynx and oesophagus cancers. The cervix uteri cancer rate was substantially higher in India than in any other study region. The stomach cancer rate among Singapore Indians was twice that of any other group.

Comparison of age-specific incidence rates

Distinct patterns were apparent in the age-specific curves according to cancer (Figure 2, panels A and B).

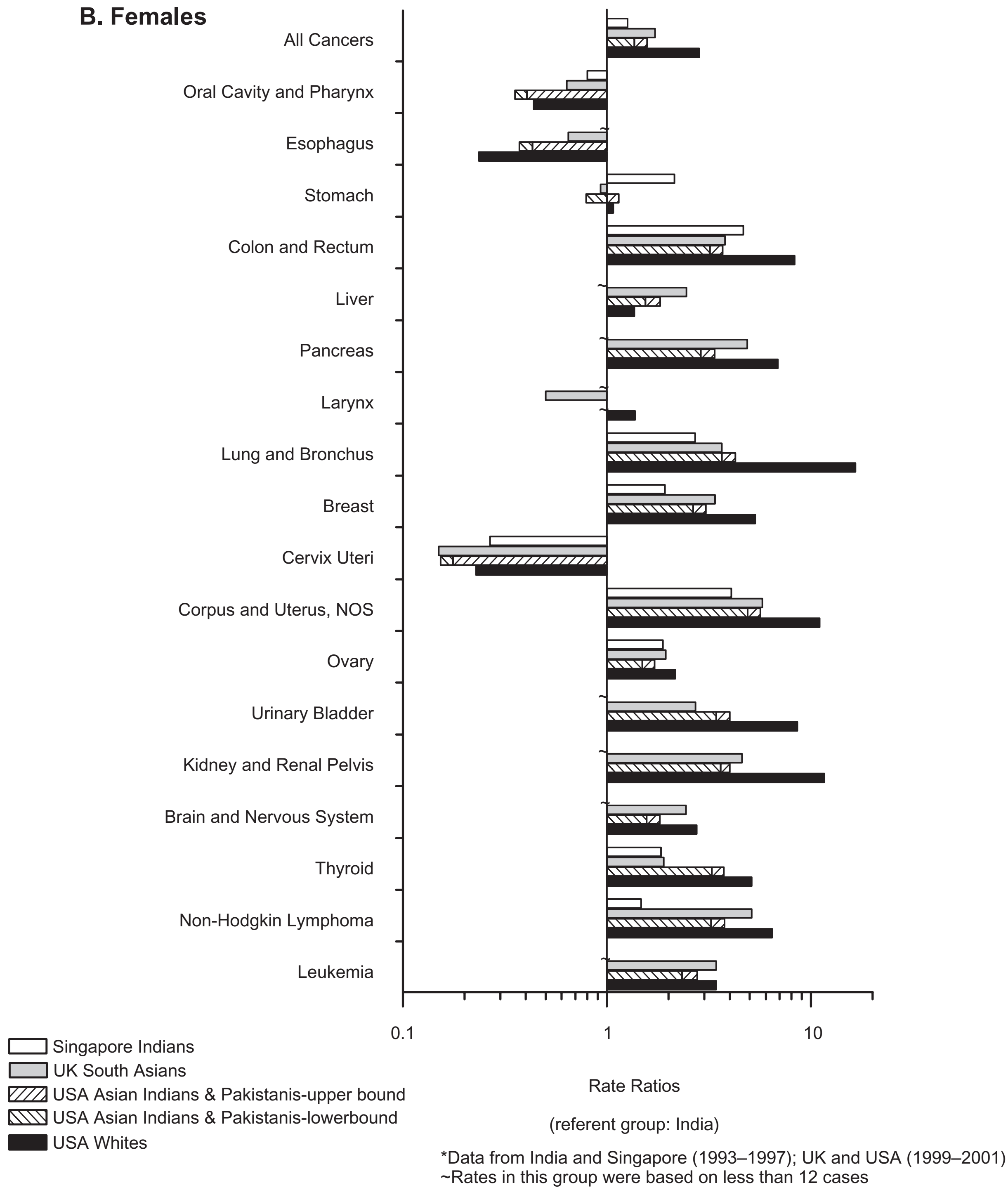

Figure 1 Continued

In the first pattern, US whites had the highest rates across most age groups, with more pronounced differences at older ages: for all cancers combined and cancers of the colorectum, pancreas, lung/bronchus, female breast, corpus uteri, bladder and kidney and renal pelvis. The relative difference at older ages for lung cancer was greater among females than males. Also, the divergence in female breast cancer rates is clearly evident in post-menopausal but not pre-menopausal women. In contrast, for prostate and thyroid cancers, the variation in rates between populations was most pronounced at the middle ages.

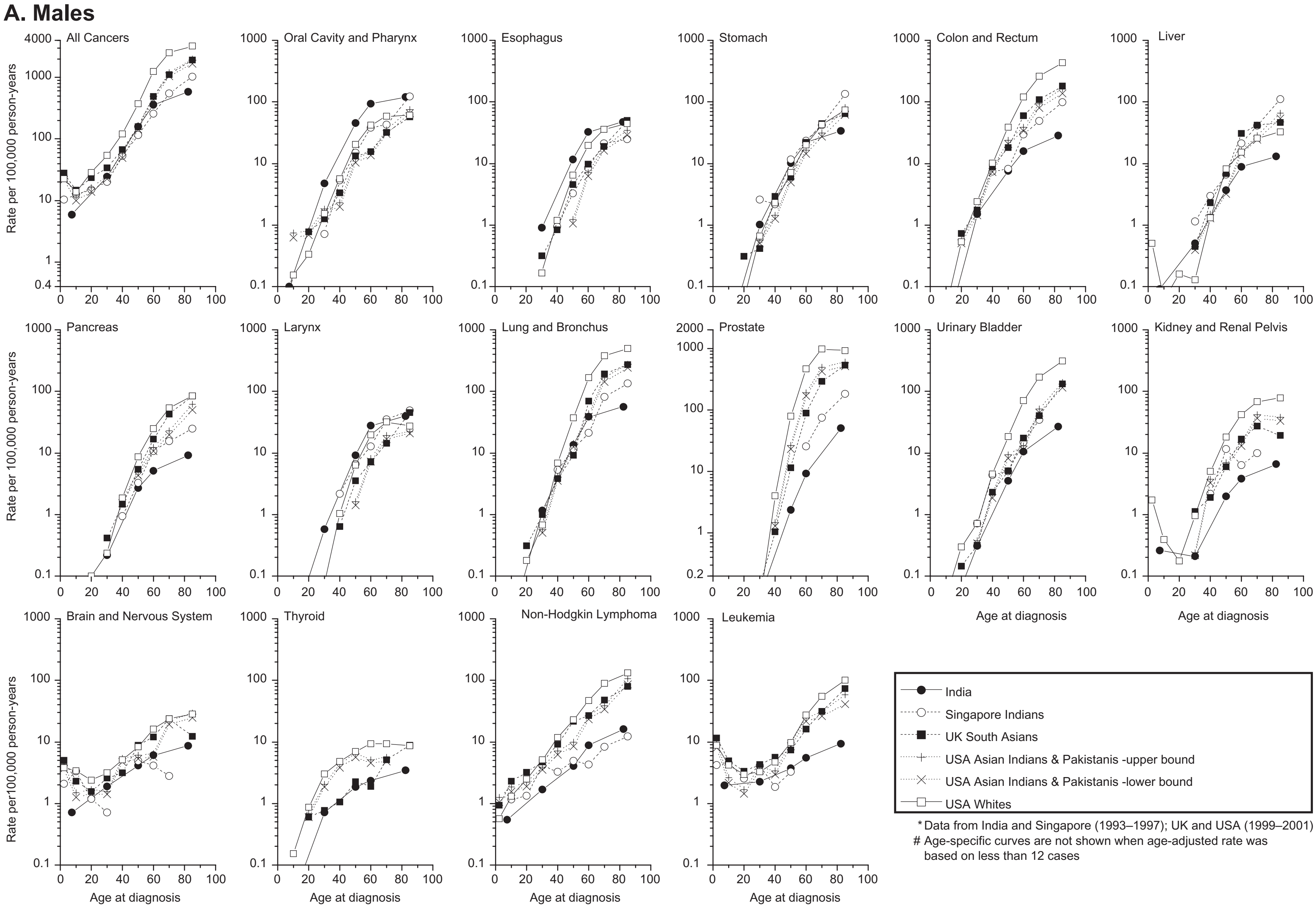

Figure 2 Age-specific incidence rates for specific cancers in India, three Indian groups (residing in Singapore, UK or US) and US whites: **A**. Males; **B**. Females

B. Females

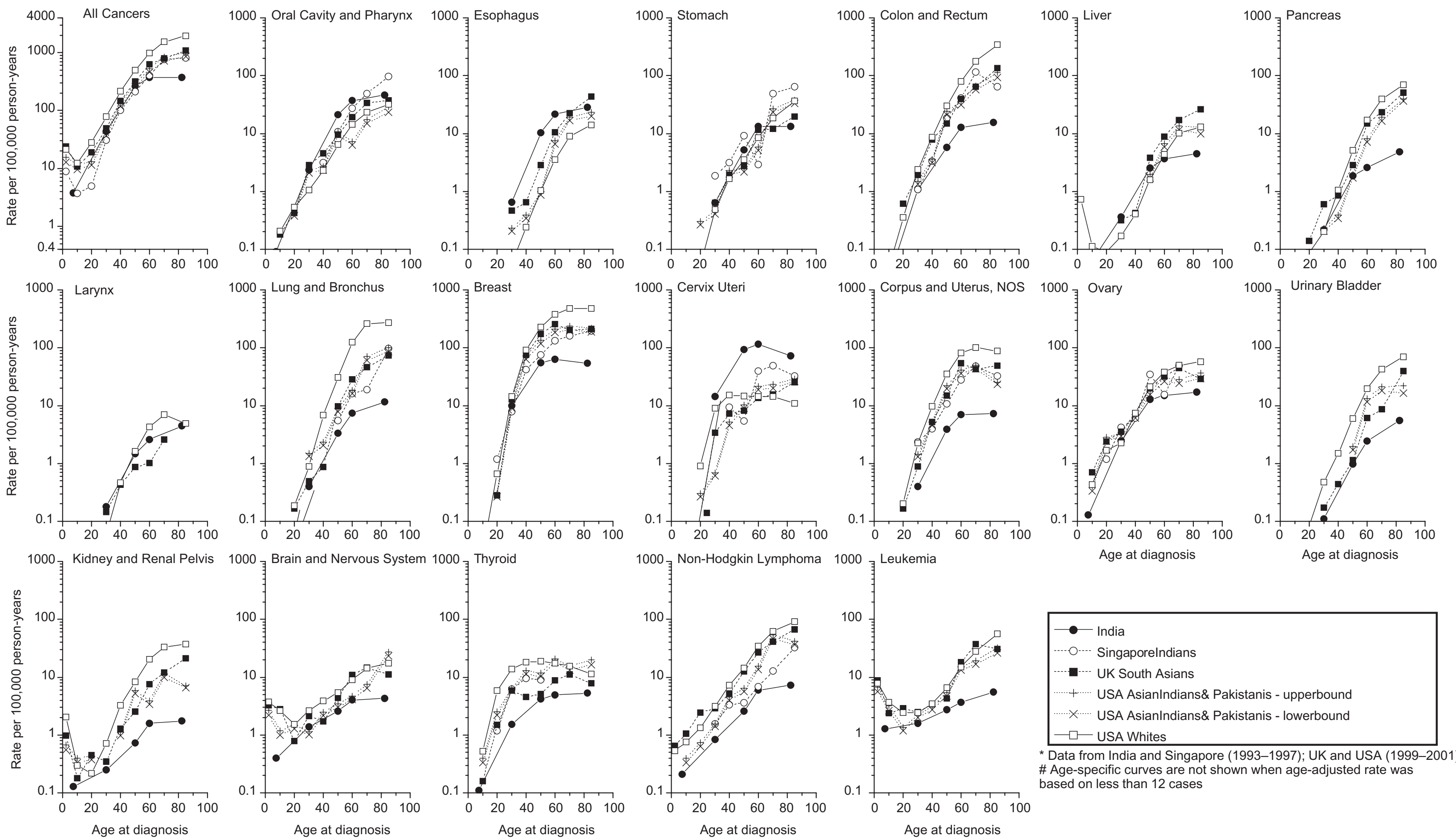

Figure 2 Continued

Incidence rate patterns among groups remained relatively constant across all ages for NHL and brain cancers.

In a second pattern, rates were highest in India across most or all age groups for cancers of the oral cavity/pharynx, oesophagus and cervix uteri. In a third pattern, remarkably little geographic variability was apparent across groups: cancers of the stomach, larynx and ovary. Childhood peaks for kidney cancer, brain cancer and leukaemia are not apparent in India due to the availability of only grouped data for ages 0–14 years.

In both the RR bar graphs and the age-specific curves, the US AIP rates varied little according to estimates of population at risk (upper and lower bounds) and had little impact on the overall patterns observed.

Discussion

Among both males and females, the incidence rate for all cancers combined and cancers of the prostate, female breast, corpus uteri, colorectum, lung, urinary bladder and kidney, were lowest in India, highest among US whites, and intermediate among Indians in Singapore, the UK and the US. In contrast, rates were highest in India for oral cavity/pharynx, oesophagus, male larynx and cervix uteri cancers. Stomach and ovarian cancer rates varied little geographically, except that stomach cancer rates were high in Singapore Indians and the ovarian cancer rate was low in India.

The low rates in India compared with US whites and SA in UK and US may be due partially to underdiagnosis but may also be due to lifestyle and environmental factors. In India there exists wide variability in dietary patterns, physical activity levels and environmental exposures.[7] There are unique aspects of the diet,[28] ranging from high intakes of the spice turmeric, containing curcumin with anticarcinogenic properties,[29] to the common practice of vegetarianism, primarily for religious and not health reasons. The higher cancer rates overall observed among Indians living in the UK and US may be due to acculturation, as seen for other ethnic groups when they migrate, or to economic changes. This pattern has been noted for other US Asian groups;[5,6] for example, colorectal cancer rates among Japanese Americans approached rates among US whites[6] within one generation. This is already apparent for breast cancer incidence, the leading cancer in all Indian-SA women outside India;[13] this has stimulated systematic investigation of environmental factors such as the potential protective effect of SA diets against breast cancer.[14] Moreover, increases over the past several decades in breast cancer incidence have been observed in Bangalore, Chennai, Mumbai, Nagpur and Pune registries. These changes may reflect the effects of rapid transition towards industrialization and urbanization as well as adoption of semi-Western diets and lifestyles, including childbearing patterns, among higher socioeconomic urban subgroups.[30–32]

The higher incidence of cancers of the oral cavity, oesophagus and larynx in India, compared with SA groups outside India, probably is due to the use of multiple tobacco products, which is less common in the West. The majority of these cases in India, for example, have been attributable to chewing tobacco and the common practice of chewing betel quids (with and without tobacco).[33–37] Decreases in oral cancer in Mumbai have been reported.[38] In contrast, lung cancer rates, which are quite modest among Indians everywhere, still are higher in the UK and the US than in India, likely reflecting the differences in the prevalence of cigarette smoking.

Cervical cancer, the most common cancer in women in India, is causally associated with exposure to the human papilloma virus (HPV).[39,40] The age-adjusted incidence rates of cervical cancer among the different registries in India varied from 10.9 in Trivandrum to 65.5 in Ambillikai, a rural area. High literacy rates and health-care awareness among women in Trivandrum may have contributed to the lower rates.[41] Declines in cervical cancer rates have been observed in Mumbai and Chennai, the long-standing urban registries. The lower rates in Singapore Indian, UK SA and US AIP females similarly may reflect better health knowledge and access to screening, including pap smears, and appropriate management of cervical pre-cancer.

There was, however, little geographic variability in stomach cancer incidence, with low rates among all studied groups except Singapore Indians. This trend is opposite to what has been observed among US Japanese, Korean and Chinese migrants, where stomach cancer rates were lower in these groups compared with their native countries.[42,43] This may be related to the lower intake in the UK and the US of salty and nitrite/nitrate-rich foods such as preserved foods and meats, which are associated with gastric cancer incidence.[44–46] In contrast, Indians may increase their intake of local foods when they move to Singapore. In Singapore, the stomach cancer rates among Indians were intermediate between those of the Chinese (higher) and Malays (lower).[3]

Ovarian cancer rates varied little among Indians in Singapore, the UK and US and US whites. The rate for India as a whole was notably lower, although within India, the rates vary considerably, with a strong urban/rural gradient.[3] In fact, rates in the urban areas were similar to those among Indians in Singapore, the UK and the US. The patterns suggest possible underdiagnosis as well as a potential role of behavioural or environmental factors.

Prostate cancer rates varied more than 20-fold between India and US whites, with a pronounced gradient among Indians in Singapore, the UK and the US. All the registry-specific rates in India[3] were substantially lower than the rates among Indians in these other areas. As yet, prostate cancer rates have not increased in Mumbai,[47] in contrast to many other areas of the world.[48] Part of the variation is likely due to diagnosis and screening practices, but the roles of environmental, behavioural and genetic factors are unknown.

This study adds valuable new information on cancer incidence rates among US AIP, which until now was unavailable. The US AIP population is large (1.84 million), and the numbers of cases were sufficient to yield fairly precise estimates of the rates. High-quality data from SEER were supplemented with data from registries of states with large US AIP populations, including California, New York, New Jersey, Texas, Florida, Pennsylvania and Michigan. The newly derived incidence rates among UK SA were obtained from 10 regional cancer registries and are nationally representative. Data on Indians in Singapore were obtained from Cancer Incidence in Five Continents (Volume VIII) and thus have met quality standards. As cancer rates in the various regions of India vary considerably, we used the Globocan national rates for India that were estimated using data from the nine population-based registries contributing to Cancer Incidence in Five Continents (Volume VIII), taking into account regional and urban/rural patterns. This should be preferable to using the simple aggregated data, which would be dominated by the data for Mumbai (Bombay), the largest Indian registry. Rates were higher for all cancers combined, colon and rectum, lung and bronchus, breast, prostate and many other cancers in Mumbai than in the Globocan total India estimates, and considerably lower for cervix uteri cancer. Although the Mumbai population includes native-Mumbai residents and people who have moved there from all over India, in varying proportions, they are not a nationally representative sample. The Globocan estimates thus are more likely to reflect the national Indian experience. Furthermore, the SA in Singapore, the UK and the US have migrated from different parts of India, so it may be more appropriate to use a nationally representative sample for India until additional information on cancer incidence among the specific migrant groups in these countries is available.

Due to variability in coding ethnicity among the national cancer registry programs, we were unable to identify SA/Indian cancer cases in regions outside India using the same methodology. Though we use data for India as our referent group, we were not able to limit our analyses in the UK and US to only Indians: we had to include persons of other SA origin because the cases were not identified separately in each of the countries. However, Indians are the majority of the US AIP population (92% Indian and 8% Pakistani) and the largest UK SA group (46% Indian, 31% Pakistani, 12% Bangladeshi and 11% other). In the UK data, we cannot exclude the potential for inaccurate identification of SA cases (numerator) with use of the name algorithm; however, the reliability of this algorithm was shown to be high.[18] Even though the cancer rates were estimated by unavoidably dissimilar methods in the various countries, the observed cancer patterns are striking and may provide impetus for various cancer registries to collect more detailed information by ethnicities. As well, we were unable to present a comparison of cancer incidence rates for exactly the same time periods across groups (India and Singapore: 1993–97; UK SA, US AIP, US whites: 1999–2001). The geographic variations observed, however, appear substantially larger than what could be accounted for by temporal changes.

While these ecologic data cannot be used to assess causal associations, they can stimulate further, more focused studies into cancer aetiology. Indians in Singapore, the UK and the US have come from different regions of India. Future investigations should compare rates among the same Indian ethnic subgroup residing in India and in other geographic regions. Detailed studies could also compare lifestyles, changes in rates between first- and second-generation groups, variability in stage of diagnosis between populations, and the impact of screening on cancer rates. Cancer registration in India has expanded greatly since it was established in Mumbai in 1964 by the Indian Cancer Society. Since 1982, the Indian Council for Medical Research has made considerable efforts to expand the national cancer registry program, which now includes 18% of the urban and 1% of the rural populations. Although the areas and populations covered by the registries are small, nonetheless the data have helped us to understand cancer patterns in India and allowed the development of Cancer Atlas.[49] This information is crucial for developing cancer-control programs in India as well as for international comparisons.

In this study, we evaluated cancer incidence among Asian Indian populations living in diverse geographic locales. The total cancer incidence rates were lower in India than among Indians in Singapore, the UK and the US, as were the rates for many cancers, with a strong gradient for several. Rates for certain cancers were higher in India than among Indians living elsewhere. These observations suggest the role of environmental and lifestyle factors as well as possible diagnostic and screening practice differences. Future research should focus on these patterns to advance our understanding of cancer aetiology and to develop the means of prevention.

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Conflict of interest: None declared.

KEY MESSAGES

- International migration studies, where the cancer incidence rates of the same genetic population residing in different geographic regions are compared, have highlighted the role of environment and lifestyle in cancer aetiology.
- Incidence rates in India for all cancers combined and 19 specific cancers were compared with those among South Asian groups residing in Singapore, UK, and US, and among US whites.
- Overall cancer incidence rates and rates of colorectal, prostate, thyroid, pancreas, lung, and breast cancer and NHL were lowest in India and highest in US whites, with intermediary rates observed in South Asians residing outside India (Singapore, UK and US).
- In contrast, incidence rates were highest in India for cancers of the oral cavity, oesophagus, larynx and cervix uteri compared with other geographic regions.
- UK and US SA appear to adopt cancer incidence patterns of their host country. Overall differences in cancer rates may be attributed to variations in lifestyle factors, including tobacco use, diet and infection, as well as health care and knowledge.

References

1 Parkin DM, Bray F, Ferlay J, Pisani P. Global cancer statistics, 2002. *CA Cancer J Clin* 2005;**55:**74–108.

2 Ferlay J, Bray F, Pisani P, Parkin DM. GLOBOCAN 2002: Cancer Incidence, Mortality and Prevalence Worldwide. IARC Cancer Base No. 5, version 2.0. Lyon: IARC Press, 2004.

3 Parkin DM, Whelan S, Ferlay J, Teppo L, Thomas DB. Cancer Incidence in Five Continents Volume VIII. IARC Scientific Publication No. 155. Lyon, France: International Agency for Research on Cancer (IARC) Press, 2002.

4 Parkin DM. International variation. *Oncogene* 2004;**23:**6329–40.

5 Yu H, Harris RE, Gao YT, Gao R, Wynder EL. Comparative epidemiology of cancers of the colon, rectum, prostate and breast in Shanghai, China vs the United States. *Int J Epidemiol* 1991;**20:**76–81.

6 Dunn JE. Cancer epidemiology in populations of the United States–with emphasis on Hawaii and California–and Japan. *Cancer Res* 1975;**35:**3240–45.

7 Rastogi T, Hildesheim A, Sinha R. Opportunities for cancer epidemiology in developing countries. *Nat Rev Cancer* 2004;**4:**909–17.

8 US Census Bureau. Census 2000 Summary File 2 (SF2) 100-Percent Data; using American FactFinder QuickTables. Available at http://factfinder.census.gov Accessed June 10, 2004. US Department of Commerce Economics and Statistics Administration UCB (ed.), 2004.

9 Reeves T, Bennet C. We the people: Asians in the United States, Census 2000 Special Reports. CENSR-17, 2004, U.S. Department of Commerce, Economics and Statistics Administration, U.S. Census Bureau. Available at http://www.census.gov/and/journals/cancer.pdf.

10 UK Census. Available at http://www.statistics.gov.uk/census/default.asp, Accessed April 8, 2004.

11 Lin SS, Clarke CA, Prehn AW, Glaser SL, West DW, O'Malley CD. Survival differences among Asian subpopulations in the United States after prostate, colorectal, breast, and cervical carcinomas. *Cancer* 2002;**94:**1175–82.

12 Jain RV, Mills PK, Parikh-Patel A. Cancer incidence in the south Asian population of California, 1988–2000. *J Carcinog* 2005;**4:**21–34.

13 Parikh-Patel A, Mills PK, Jain RV. Breast cancer survival among South Asian women in California (United States). *Cancer Causes Control* 2006;**17:**267–72.

14 dos Santos Silva I, Mangtani P, McCormack V, Bhakta D, Sevak L, McMichael A. Lifelong vegetarianism and risk of breast cancer: a population-based case-control study among South Asian migrant women living in England. *Int J Cancer* 2002;**99:**238–44.

15 Harding S, Rosato M. Cancer incidence among first generation Scottish, Irish, West Indian and South Asian migrants living in England and Wales. *Ethn Health* 1999;**4:**83–92.

16 Moles DR, Fedele S, Speight PM, Porter SR. The unclear role of ethnicity in health inequalities: the scenario of oral cancer incidence and survival in the British South Asian population. *Oral Oncol* 2007;**43:**831–34.

17 Keegan TH, Gomez SL, Clarke CA, Chan JK, Glaser SL. Recent trends in breast cancer incidence among 6 Asian groups in the Greater Bay Area of Northern California. *Int J Cancer* 2007;**120:**1324–29.

18 Nanchahal K, Mangtani P, Alston M, dos Santos Silva I. Development and validation of a computerized South Asian Names and Group Recognition Algorithm (SANGRA) for use in British health-related studies. *J Public Health Med* 2001;**23:**278–85.

19 National Cancer Institute, Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Incidence–SEER 11 Regs + AK Public-Use, Nov 2003 Sub (1973–2001 varying). Released April 2004, based on the November 2003 submission. Available at http://www.seer.cancer.gov. Accessed June 10, 2004. National Cancer Institute, DCCPS, Surveillance Research Program, Cancer Statistics Branch.

20 California Cancer Registry. Analysis Created in July 2004 using Statewide Cancer Database, Released May 2004. California Department of Health Services, Cancer Control Branch, Cancer Surveillance Program, Sacramento, CA, 2004.

21 New York State Cancer Registry. Analysis Created in July 2004. Albany, NY: New York State Department of Health, 2004.

22 New Jersey State Cancer Registry. Analysis created in August 2004 using SEER*Stat database: NJ Incidence by Race (WBO) & County 1979–2001–(Use custom Race), V9.1, Bladder In situ recoded as Inv (Created 2/6/04 by XN). Trenton, NJ: New Jersey Department of Health and Senior Services, Cancer Epidemiology Services, 2004.

23 Texas Cancer Registry. Analysis created in July 2004 using SEER*Stat database: Texas Cancer Registry 1995–2001 Incidence File as of 12/02/2003. Austin, TX: Texas Department of State Health Services, Cancer Epidemiology and Surveillance Program, 2004.

24 Florida Cancer Data System. Analysis Created in July 2003. Miami, FL: University of Miami, Miller School of Medicine, 2004.

25 Pennsylvania Department of Health. 'Annual Cancer Incidence Files, 1999–2001', Released July 2004. Harrisburg, PA: Pennsylvania Cancer Registry, 2004.

26 Michigan Resident Cancer Incidence File. Includes cases diagnosed in 1999–2001 and processed by the Michigan Department of Community Health, Vital Records and Health Data Development Section by 14 November 2003; 2004.

27 Miettinen OS, Nurminen M. Comparative analysis of two rates. *Stat Med* 1985;**4:**213–26.

28 Sinha R, Anderson DE, McDonald SS, Greenwald P. Cancer risk and diet in India. *J Postgrad Med* 2003;**49:**222–28.

29 Weber WM, Hunsaker LA, Abcouwer SF, Deck LM, Vander Jagt DL. Anti-oxidant activities of curcumin and related enones. *Bioorg Med Chem* 2005;**13:**3811–12.

30 Yeole BB, Kurkure AB. An epidemiological assessment of increasing incidence and trends in breast cancer in Mumbai and other sites in India, during the last two decades. *Asian Pac J Cancer Prev* 2003;**4:**51–56.

31 Murthy NS, Agarwal UK, Chaudhry K, Saxena S. A study on time trends in incidence of breast cancer–Indian scenario. *Eur J Cancer Care* 2006;**16:**185–86.

32 Colditz GA. Epidemiology and prevention of breast cancer. *Cancer Epidemiol Biomarkers Prev* 2005;**14:**768–72.

33 Moore SR, Johnson NW, Pierce AM, Wilson DF. The epidemiology of mouth cancer: a review of global incidence. *Oral Dis* 2000;**6:**65–74.

34 Babu KG. Oral cancers in India. *Semin Oncol* 2001;**28:**169–73.

35 Sankaranarayanan R. Oral cancer in India: an epidemiologic and clinical review. *Oral Surg Oral Med Oral Pathol* 1990;**69:**325–30.

36 Rahman M, Sakamoto J, Fukui T. Calculation of population attributable risk for bidi smoking and oral cancer in south Asia. *Prev Med* 2005;**40:**510–14.

37 Jacob BJ, Straif K, Thomas G, Ramadas K, Mathew B, Zhang ZF *et al*. Betel quid without tobacco as a risk factor for oral precancers. *Oral Oncol* 2004;**40:**697–704.

38 Sunny L, Yeole BB, Hakama M *et al*. Oral cancer in Mumbai India: a fifteen years perspective with respect to incidence trend and cumulative risk. *Asian Pac J Cancer Prev* 2004;**5:**294–300.

39 Bekkers RL, Massuger LF, Bulten J, Melchers WJ. Epidemiological and clinical aspects of human papilloma virus detection in the prevention of cervical cancer. *Rev Med Virol* 2004;**14:**95–105.

40 Franco EL, Schlecht NF, Saslow D. The epidemiology of cervical cancer. *Cancer J* 2003;**9:**348–59.

41 Murthy NS, Chaudhry K, Saxena S. Trends in cervical cancer incidence–Indian scenario. *Eur J Cancer Prev* 2005;**14:**513–18.

42 Kamineni A, Williams MA, Schwartz SM, Cook LS, Weiss NS. The incidence of gastric carcinoma in Asian

migrants to the United States and their descendants. *Cancer Causes Control* 1999;**10:**77–83.

43 Kolonel LN, Nomura AM, Hirohata T, Hankin JH, Hinds MW. Association of diet and place of birth with stomach cancer incidence in Hawaii Japanese and Caucasians. *Am J Clin Nutr* 1981;**34:**2478–85.

44 Tsugane S, Sasazuki S, Kobayashi M, Sasaki S. Salt and salted food intake and subsequent risk of gastric cancer among middle-aged Japanese men and women. *Br J Cancer* 2004;**90:**128–34.

45 Lee JK, Park BJ, Yoo KY, Ahn YO. Dietary factors and stomach cancer: a case-control study in Korea. *Int J Epidemiol* 1995;**24:**33–41.

46 Ji BT, Chow WH, Yang G *et al*. Dietary habits and stomach cancer in Shanghai, China. *Int J Cancer* 1998;**76:**659–64.

47 Sunny L, Yeole BB, Kurkure AP *et al*. Cumulative risk trends in prostate cancer incidence in Mumbai, India. *Asian Pac J Cancer Prev* 2004;**5:**401–5.

48 Hsing AW, Devesa SS. Trends and patterns of prostate cancer: what do they suggest? *Epidemiol Rev* 2001;**23:**3–13.

49 Nandakumar A, Gupta PC, Gangadharan P, Visweswara RN (eds). Development of an atlas of cancer in India. First all India Report: 2001–2002. Bangalore, India: National Cancer Registry Program (ICMR), 2004.

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Commentary: Cancer incidence among Asian Indians in India and abroad

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Increased migration is a reality in the world and migration has social, cultural, political and economic consequences impacting the health of migrant communities. Studying the impact of migration on health in general, on cancer and other chronic diseases in particular, can offer important information on the role of environmental and ethnic factors in disease aetiology. Although there are some difficulties in differentiating the influences of environmental risk factors from those of genetic racial factors, the information can be used to develop aetiological studies and cancer control interventions. Studies on cancer incidence rates among Japanese in Japan and Japanese and Caucasians in the US revealed that the risks of cancers of the prostate, corpus uteri, colon, thyroid, breast, ovary and testis among the Japanese migrants and their descendants were elevated, whereas those of the stomach, oesophagus and cervix uteri were decreased among them in the US as compared with the Japanese in Japan.[1] Although there have been past studies among Hispanics,[2] Japanese,[1,3] Chinese, Filipino, Vietnamese, Korean ethnicities[3] and European migrants[4] in the US, cancer incidence among Asian Indians living in other countries has not been widely studied. The article by Rastogi and colleagues[5] describing the cancer profile among Indians in India and persons of Indian origin in Singapore, UK and the US is an important addition to the literature on changing cancer risks and patterns among migrants.

Rastogi and colleagues used the GLOBOCAN estimates for India for 1993–1997 and cancer incidence data from the Singapore Cancer Registry for 1993–97; the data for 1999–2001 for the UK were derived from the National Cancer Intelligence Centre (NCIC) and that for the US from Surveillance, Epidemiology and End Results (SEER) programme for 1999–2001. Thus the comparison periods used are different, which is basically due to the non-availability of reliable data for corresponding periods. Inspite of the data limitations due to the possible discrepancies in the completeness of the counts of cancer cases and the population estimates for South Asian Indians in the UK and US and use of estimated data for Indians in India, the findings are consistent with existing knowledge about the importance of potentially modifiable environmental and behavioural determinants of cancer risk. The comparison brings forth the importance of cancer registration, early detection programmes, development and access to health services and health care coverage in cancer control and prevention. The higher rates of prostate,

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